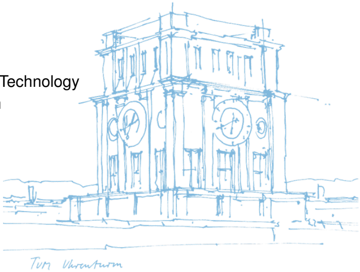
Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

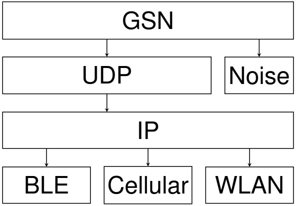
Dan Bachar, David Guzman

Monday 18th May, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

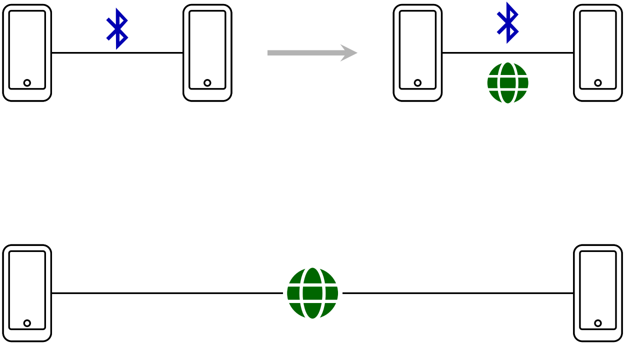


Architecture



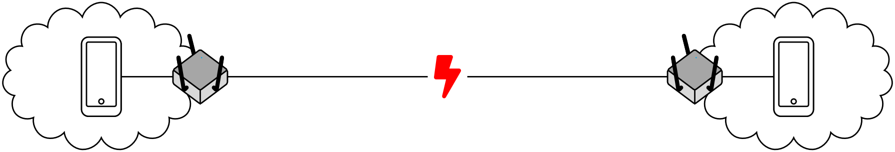
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

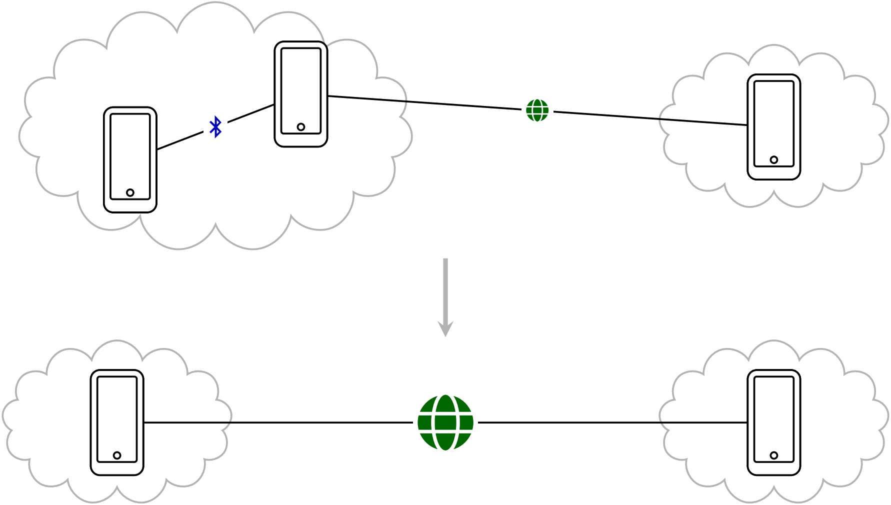


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

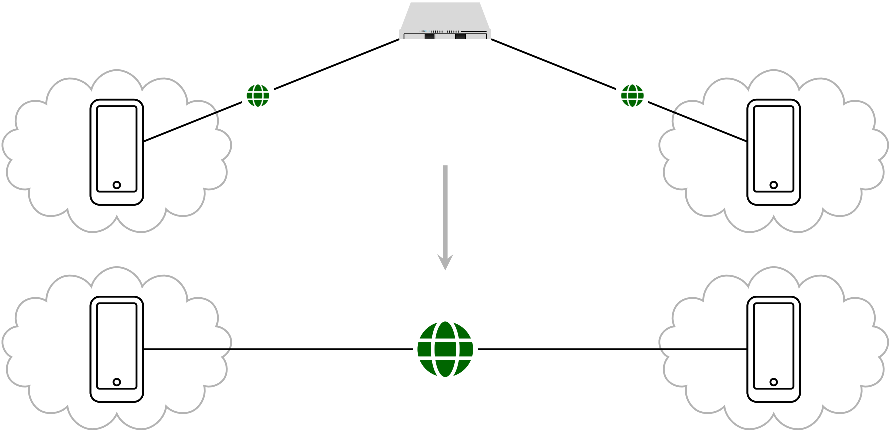
NAT



Well-connected friends



Rendezvous Server



Since last weekly

- Migrated BLE communincation layer from two separate central/peripheral plugins to one `grassroots_bluetooth_layer`: unified event model, shared discovery UUID, post-connect ANNOUNCE identity, and BLE role toggle.
- Reworked UDX into a dual-stack Internet transport: IPv4/IPv6 sockets, multi-candidate addresses, live network/public-address refresh, and best candidate-pair selection.
- Aligned rendezvous/NAT traversal with GLP Networking API doc: two-sided punch, RV fan-out (A and B try each others' RV servers when the connection breaks), friends-of-friends mediation (friends can facilitate rendezvous for mutual friends).
- Persist friend data in encrypted local storage so that connections can be initiated post restart.
- Added Noise XX handshake for secure channel encryption: messaging and signaling with Noise XX across clients and rendezvous anchors.
- Hardened delivery over UDX/BLE: ACKs after fragment reassembly, increase max UDX payload size for photos, and stale-session/socket recovery (app loses connections when backgrounded)

Mobility model: city-scale BLE relay

Protocol question

- Given real city encounters, when does sealed BLE relay deliver?
- Compare direct, friend-only, and open relay policies.
- Existing mobility traces give contacts; we measure protocol outcomes.

What we track

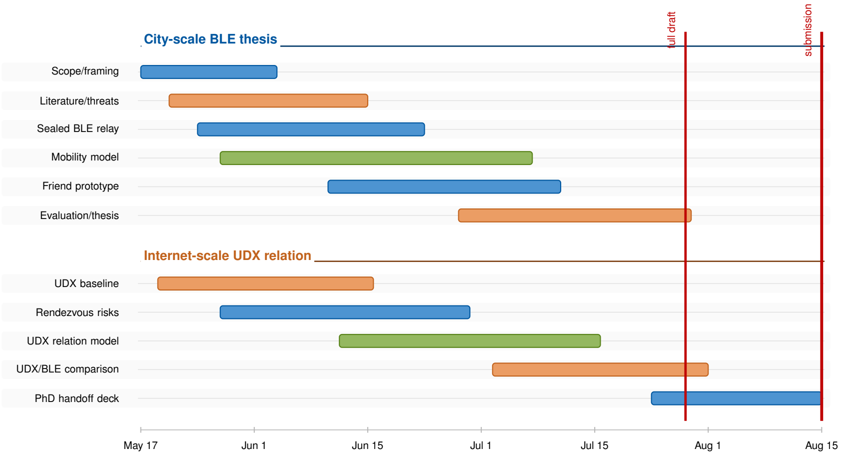
- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost/risk: relay bytes, duplicates, queues, scan/advertise time, battery drain.

How

- Use existing BLE/contact models as baselines or priors.
- Calibrate with encrypted local trace logs from our Flutter stack.
- Replay traces in simulator with relay policy, TTL, ACKs, and queues.

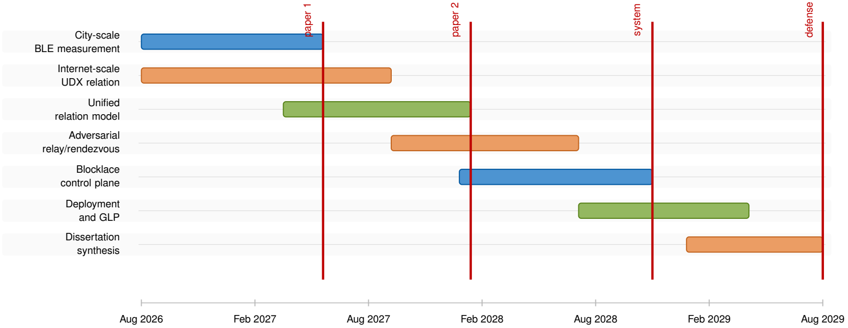
Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

Master gantt: thesis plus UDX track



- Thesis core: BLE mobility and secure local relay; Keidar–Shapiro thread: keep UDX as an Internet-scale relation model and comparison.

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.